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Paleomagnetism and Neogene Clockwise Rotation of the Northern Cady Mountains, Mojave Desert of Southern California

BRUCE J. MACFADDEN

Florida Museum of Natural History, University of Florida, Gainesville

MICHAEL O. WOODBURNE

Department of Geological Sciences, University of California, Riverside

NEIL D. OPDYKE

Department of Geology, University of Florida, Gainesville

The early to middle Miocene Hector Formation of the northern Cady Mountains consists of about 400 m of syntectonic alluvial fan and volcaniclastic sediments deposited in the central Mojave desert. Paleomagnetic samples were collected from 60 sites throughout this unit. Samples from each site were treated using both alternating field and thermal demagnetization. These studies and isothermal remanent magnetization experiments indicate that the natural remanent magnetization (NRM) is complex and that characteristic directions can be carried in either high coercivity or continuous unblocking temperature spectra depending upon the sample. A positive reversals test and concordant directions of the alternating field (AF) and low and high temperature components suggest a stable magnetization. Although a unique magnetostratigraphic correlation to the time scale is difficult, the section studied may represent chrons C6B to C5C. After screening of individual components and calculation of site means using Fisher statistics, 48 of the original 60 sites were used to calculate an overall formation mean direction, i.e., declination 18.6°, inclination 45.4° (α_{95}=6.0°). Using a 20-m.y. paleomagnetic pole for North America [*Irving and Irving*, 1982], the Hector Formation has undergone 20.6° ± 7.6° (R ± ΔR) of clockwise rotation around a vertical axis. This amount of rotation is considerably less than reported for some early Miocene rocks elsewhere in the Mojave Block. Least squares regression of the 48 site mean declinations versus stratigraphic position indicates no trend throughout the section, suggesting that there was no syndepositional rotation of this unit. The late Cenozoic structural history of the Mojave Block is considerably more complex than was originally believed, and currently available paleomagnetic data indicate that many blocks within this region have experienced different degrees of rotation. There were at least two phases of tectonic rotation in this region: (1) early Miocene (about 25 Ma), probably related to the initiation of extension within the Mojave Rift; and (2) post middle Miocene-Pliocene (between about 16 Ma to the present), possibly related to movement along principal structural features such as the San Andreas fault zone. The exact timing of the younger phase of tectonic activity cannot be better constrained until paleomagnetic data become available for late Cenozoic rocks within the Mojave Block.

INTRODUCTION

Within the last two decades, there has been an increased interest in the late Cenozoic tectonic evolution of the Mojave Desert. *Garfunkel* [1974] published an important paper on the "Mojave Block," a structural region bounded to the north by the Garlock fault and to the south by the San Andreas fault. Based on field geology and theoretical models, he suggested that local structural blocks bounded by NW trending faults in this region could have undergone as much as 30° of counterclockwise rotation during this time. *Luyendyk et al.* [1980] suggested that blocks bounded by E-W trending faults within this region could have undergone up to 90° of clockwise rotation, in a manner similar to the Transverse Ranges to the south of the San Andreas fault [*Luyendyk et al.*, 1985]. Realizing that the tectonic history of the Mojave Block is complex, *Garfunkel* [1974, p. 1931] stated that "the precise magnitude and timing of the rotation can be checked by paleomagnetic studies."

Since the presentation of these regional tectonic models, there have been numerous paleomagnetic studies that test these hypotheses and this expanding data base has complemented the recent studies on the geological evolution of the Mojave Block [*Dokka*, 1986; *Dokka and Woodburne*, 1986; *Dokka et al.*, 1988]. Early paleomagnetic studies of individual blocks bounded by NW trending faults [*Burke et al.*, 1982] suggested that *Garfunkel's* [1974] model might be correct. However,

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numerous recent paleomagnetic studies [*MacFadden et al.*, 1987, 1988, 1990; *Golombek and Brown*, 1988; *Ross et al.*, 1987, 1988, 1989; *Valentine et al.*, 1988; *Wells and Hillhouse*, 1986, 1987, 1989] indicate that the structural history of various blocks within the Mojave Block is considerably more complex than was originally suggested by Garfunkel's regional model. These data indicate that individual blocks have undergone independent rotational histories during the last 25 m.y. A tectonic synthesis of these data is not yet available, and more individual data points are needed before a clearer picture of the history of the Mojave Desert can be presented, particularly for the period from 15 Ma to the present.

In this paper we present new paleomagnetic data for the rotational history of the northern Cady Mountains in the eastern Mojave Block (Figure 1) as evidenced from a detailed study of the Hector Formation. Many studies of paleomagnetic rotations in the Mojave Desert have been done on isolated volcanic outcrops in which possible local deformation or non-time averaged geomagnetic fields must be carefully considered. In contrast, our study area includes a temporally well-constrained sedimentary sequence spanning about 7 m.y. The Hector Formation is of great importance because its sedimentary history is syndepositional with the major phase of detachment faulting within the Daggett Terrane that occurred in the formation of the Mojave Rift [*Dokka*, 1986; *Dokka and Woodburne*, 1986; *Dokka et al.*, 1988]. *MacFadden et al.* [1988] report preliminary paleomagnetic data indicating 32.0° ± 10.9° of clockwise rotation from the lower and middle parts of Hector Formation in the same area, but these results are based on a smaller data set than is presented here. *Ross et al.* [1988], working in the underlying early Miocene volcanic units in the southern Cady Mountains, obtained declination anomalies indicating clockwise rotation of some 57.9° ± 12.6°. (Another result of 123.9° ± 16.2° from the western Cady Mountains reported by *Ross et al.* [1988] may represent locally anomalous rotation; T. M. Ross, personal communication, 1988.)

GEOGRAPHIC AND GEOLOGIC SETTING

The Cady Mountains are located at latitude 34°45'N and longitude 116°W in the eastern part of the Mojave Desert in San Bernardino County, California (Figure 1). This region has been tectonically active during the late Tertiary and has been the subject of much recent work (for detailed discussions, see *Dokka* [1986] and *Dokka and Woodburne* [1986]). The terminology of the study region is somewhat complex, and has been termed in a physiographic/geographic context the Mojave Desert, whereas in a geological/structural context it has been variously termed the Mojave Block [*Garfunkel*, 1974], Mojave Desert block [e.g., *Dokka*, 1986], and Mojave Rift [*Dokka et al.*, 1988]. In this paper this region is termed the Mojave Block for the larger area bounded by the Garlock, San Andreas, and Death Valley faults [*Garfunkel*, 1974; *Dokka*, 1986], whereas the smaller area that underwent early Miocene E-W extension is termed the Mojave Rift [*Dokka*, 1988; *Dokka et al.*, 1988].

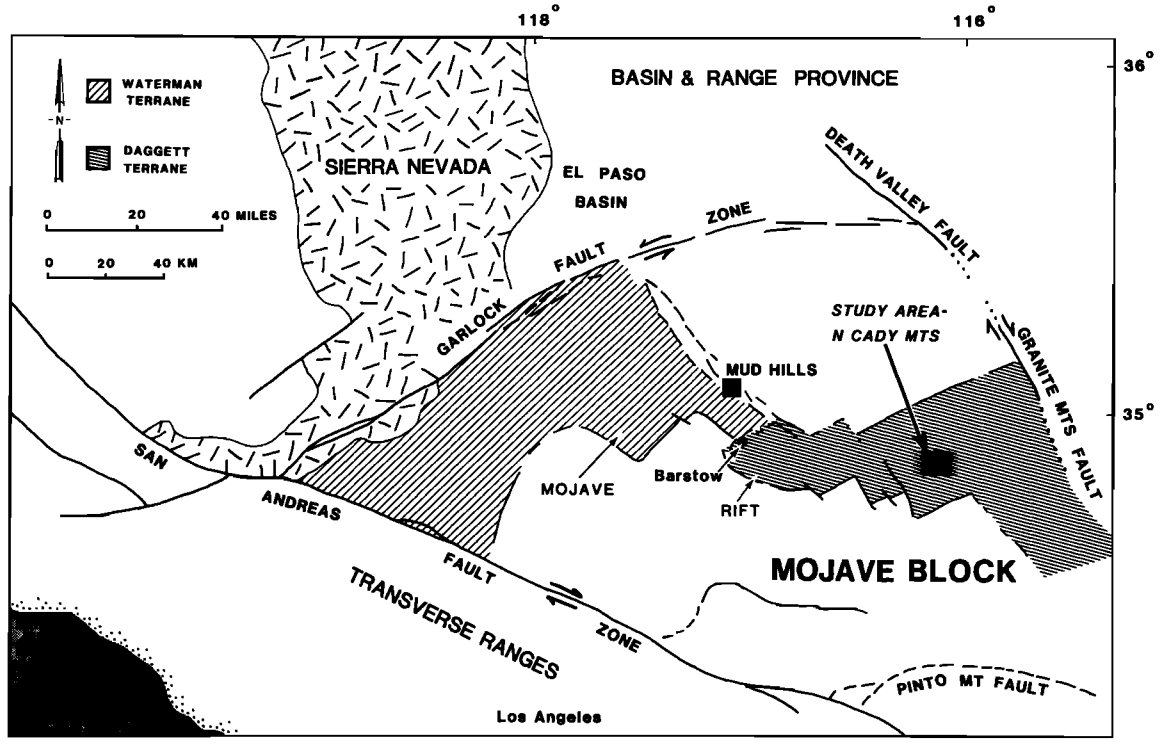


Fig. 1. Location of the study area in the northern Cady Mountains, southern California, and regional geological setting within the Mojave Block. The interpretation of geological features within the Mojave Rift is taken from *Dokka* [1986] and *Dokka and Woodburne* [1986].

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The unit studied here, the Hector Formation [*Woodburne et al.*, 1974], crops out on the flanks of the Cady Mountains. This unit consists of about 400 m of volcanic, volcaniclastic, and fluviatile sediments deposited in coalescing alluvial fans. In the type section in the southern Cady Mountains [*Woodburne et al.*, 1974] the fluviatile component of this formation is represented by predominantly coarse sands and conglomerates. However, in the northern exposures located in sections 1, 12, and 13, T. 10 N., R. 7 E. and section 36, T. 11 N., R. 6 E., U.S. Geological Survey Cady Mountains 15-minute Quadrangle, a more distal facies consists of finer-grained deposits of silts and fine sands [*Miller*, 1980], which seemed potentially amenable to paleomagnetic analysis. As such, we chose to concentrate on the section in the northern Cady Mountains (Figure 1). *Miller* [1980] presents an excellent structural and stratigraphic base for our work in this area. Relative to previous structural models, our field area is located north of the Cady fault and south of the Manix fault.

AGE OF THE HECTOR FORMATION

The age of the Hector Formation is well constrained by both isotopic and paleontological data. The longest-known

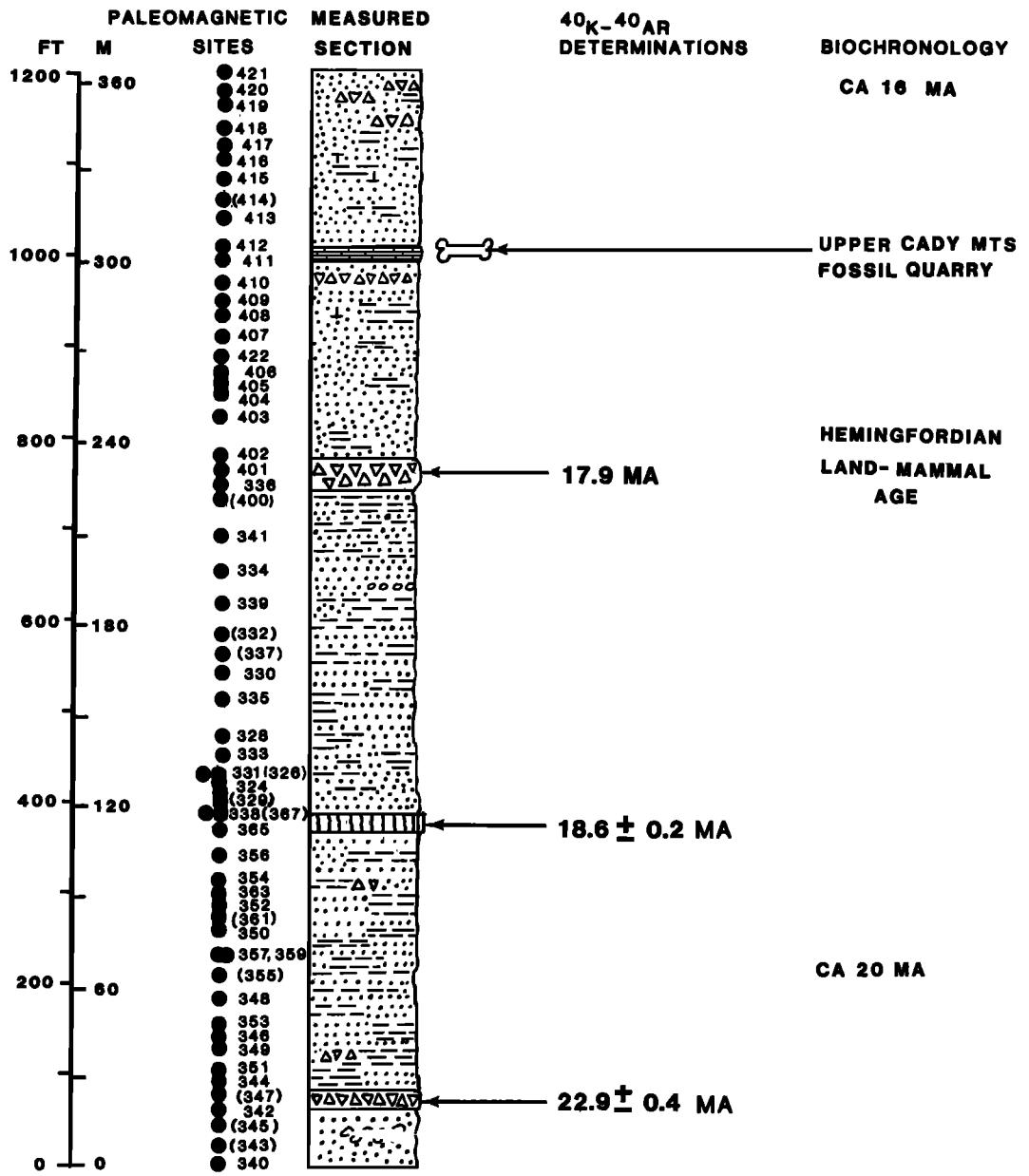


Fig. 2. Composite stratigraphic section of the Hector Formation in the northern Cady Mountains showing lithology, position of paleomagnetic sites, isotopically dated units and fossil horizons, and biochronologic age of the fossil mammals. The statistically reliable paleomagnetic sites (Class I of *Opdyke et al.* [1977]) correspond to those in Table 2. The sites in parentheses were not used in determination of rotation because they lacked reliable mean directions, although 345, 361 (both Class III), and 326 (Class II) were used for polarity determination (see Figure 7).

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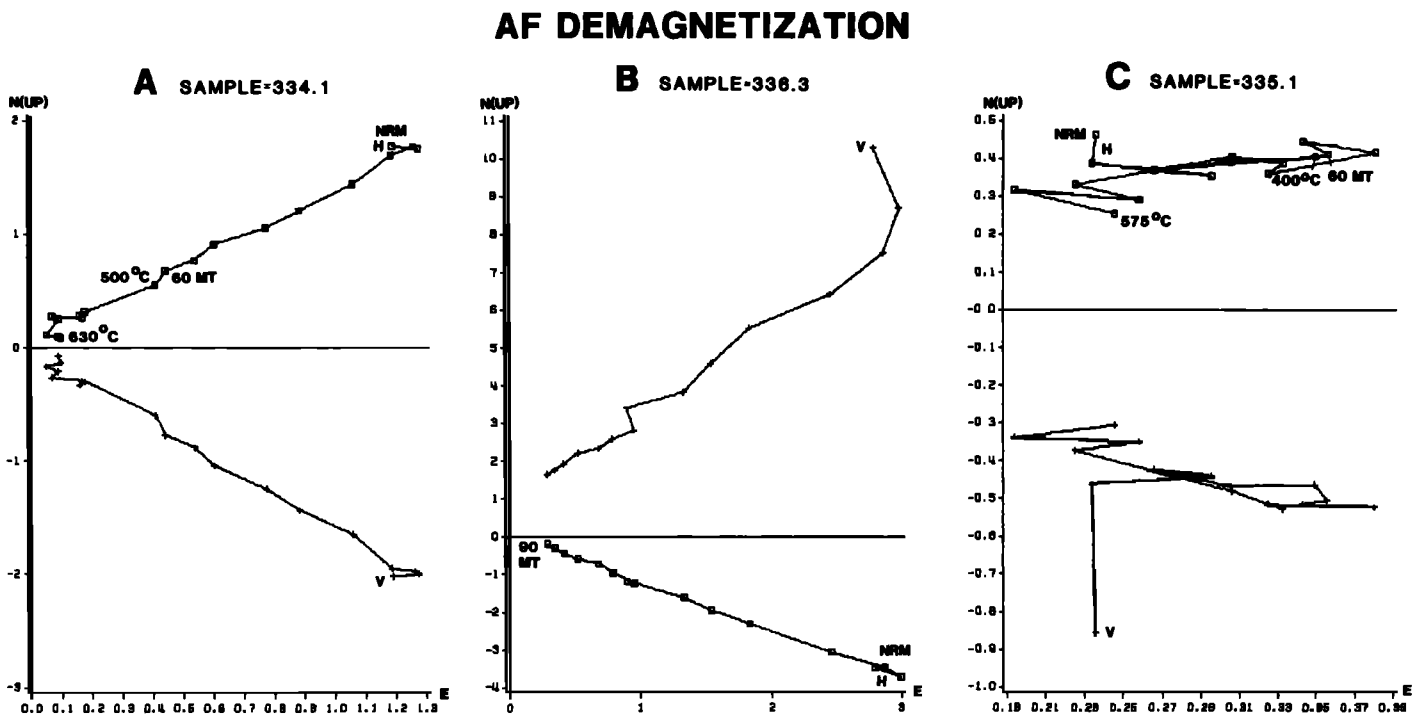


Fig. 3. Orthogonal component diagrams showing stepwise alternating field (Figures 3a-3c) and thermal (Figures 3d-3f) demagnetization vector decay for selected samples from the Hector Formation in the northern Cady Mountains.

temporal extent of this unit occurs in the northern Cady Mountains in our study area. The lower age limit of the Hector Formation is approximated by a ^{40}K-^{40}Ar age of 22.9 ± 0.4 Ma (all previously published dates presented here are corrected for the new decay constants [*Dalrymple*, 1979]) for a tuff that occurs 23 m above the base of the local section [*Miller*, 1980] (Figure 2). The age of the lower and middle parts of the formation is further constrained by superposed ^{40}K-^{40}Ar ages of 18.6 ± 0.2 Ma on a basalt at 110 m and 17.9 Ma (no analytical error is available; see *Miller* [1980]) on a rhyolitic flow between 220 and 230 m in our sampling transect (Figure 2) and *Miller's* [1980] section.

These isotopic ages are corroborated by (1) the presence of biochronologically diagnostic mammals of middle Hemingfordian land-mammal age in the lower one-third of the section, which indicate an age of about 20-18 Ma, and (2) a rich fauna of late Hemingfordian age (from the Upper Cady Mountains Quarry, Figure 2) overlying the 17.9 Ma dated rhyolite that correlates to an age range between 18 and 16 Ma [*Berggren et al.*, 1985; *Tedford*, 1987]. The age of the upper part of the section is further corroborated by new age constraints between about 19 and 16 Ma from the lower part of the Barstow Formation, located some 50 km to the west, containing a similar late Hemingfordian mammalian assemblage [*MacFadden et al.*, 1987, 1990]. In summary, based on both isotopic and paleontological data, the age constraints for the Hector Formation are between 23 and 16 Ma.

PALEOMAGNETIC ANALYSIS

Paleomagnetic samples were collected at 60 sites (i.e., individual horizons) from the volcanic and sedimentary exposures of the composite measured section of the Hector Formation in the northern Cady Mountains (Figure 2). We concentrated our sampling in this area because previous reports, and our field investigations of the other known exposures of this unit indicated that they were either too coarse or lacked a firm stratigraphic context to allow a continuous vertical transect. Sites were spaced about 8 m apart vertically, and at each, three to five separately oriented samples were collected.

In the laboratory, samples were prepared into standard 2.54 cm cubes and analyzed in a SCT two-axis cryogenic magnetometer [*Fuller et al.*, 1985] within a magnetically shielded room [*Scott and Froelich*, 1985] at the University of Florida. For each site one sample was subjected to stepwise alternating field (AF) demagnetization at peak fields between 5 and 50 mT or higher (sometimes to 100 mT) and another sample was subjected to stepwise thermal demagnetization between 100° and 650°C. In all cases these samples were subjected to between 9 and 20 steps in order to determine trends in the decay of magnetization. After removal of a low field component, presumably of viscous origin, AF demagnetization sometimes effectively resulted in stable vector decay (Figures 3a and 3b); however, in many cases a high coercivity component could not

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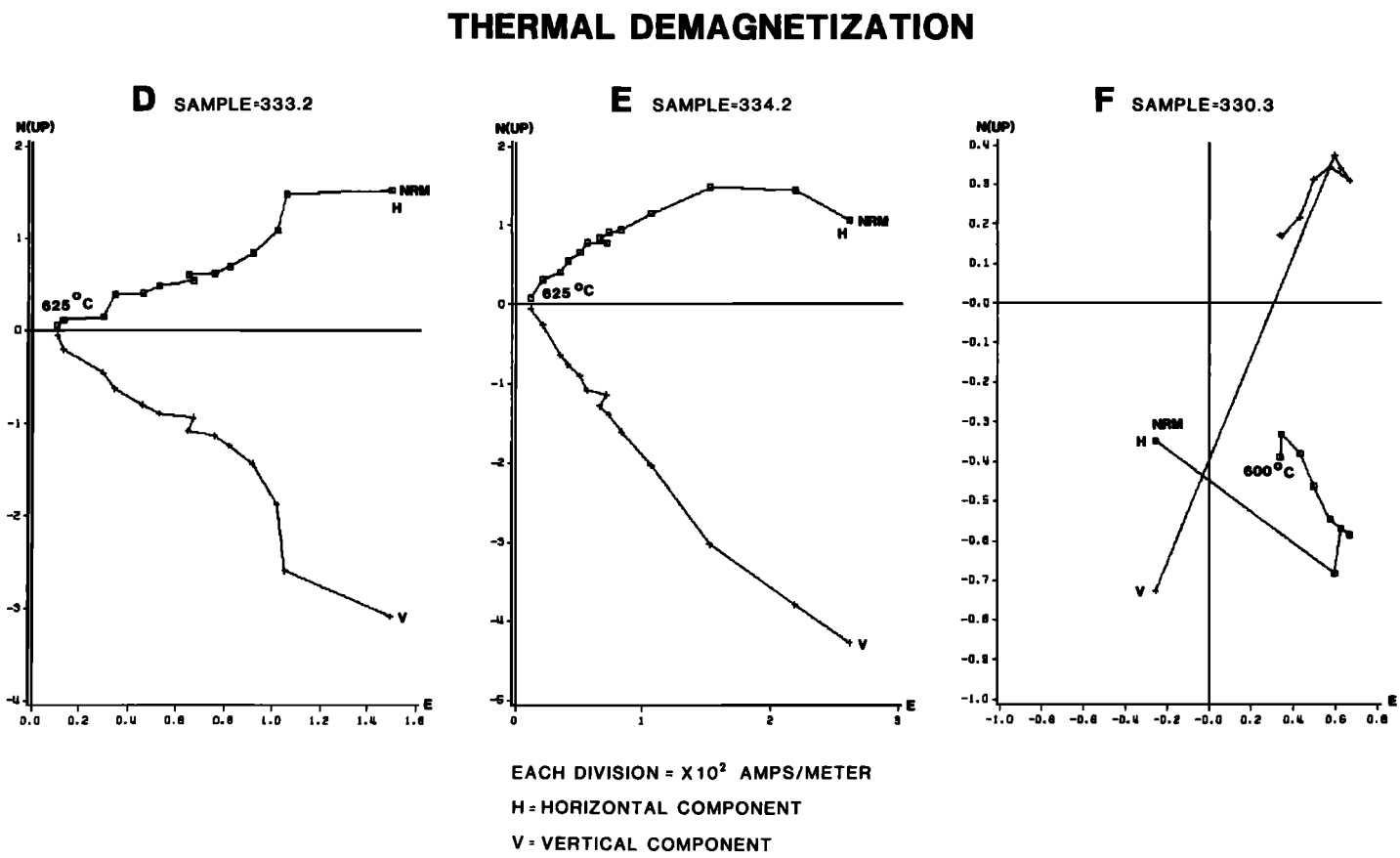


Fig. 3. (continued)

be removed using this technique (e.g., Figure 3c). In most cases thermal demagnetization effectively resulted in linear vector decay (Figures 3d-3f) after removal of low unblocking temperature components. The third sample from each site was then subjected to whichever regime seemed to most effectively isolate stable components; in most cases this turned out to be thermal demagnetization. For all samples in which a vector interpreted to be of normal polarity existed, thermal demagnetization was done even after a complete AF regime (e.g., Figures 3a and 3c).

It is obvious from the discussion above that the magnetization of the Hector Formation is complex. This is understandable because of the varied lithologies and potential sedimentary environments represented in this unit. In order to better identify the dominant magnetic minerals carrying the remanence, saturation isothermal remanent magnetization (IRM) studies were done on samples from 10 sites. Results of those experiments resulted in both low field (about 100-150 mT) saturation, probably indicating magnetite (Figure 4b), and samples in which saturation did not occur up to 500 mT, probably indicating hematite or goethite (Figures 4a, 4c, and 4d). Therefore the combined AF and thermal demagnetization regimes described above for each site seemed warranted.

After demagnetization was completed, linear segments on vector demagnetization diagrams were chosen to calculate characteristic sample directions using the least squares regression and principal component techniques [*Kirschvink*, 1980]. After removal of either low coercivity or low unblocking temperature components, characteristic directions were determined for any sample that exhibited stable behavior for the linear component resulting from either AF, low temperature thermal (usually 100°-550°C), and/or high temperature thermal demagnetization (usually 575°-650°C). In some cases, stable directions (i.e., linear trends) of opposite polarity were carried in different components from the same sample, and different samples within some sites carried antiparallel magnetizations within the same component. There is no obvious systematic correlation of magnetic behavior with general lithology. As a result of this complex magnetization, it is difficult to interpret an unambiguous magnetostratigraphy; this is discussed below. Also, our results indicate a significant declination anomaly, and hence rotation, for the Hector Formation. As indicated in Figure 5, this direction is carried in all three components. These results suggest that the magnetization resulting from these components was acquired roughly at the same time, although an inconclusive fold test

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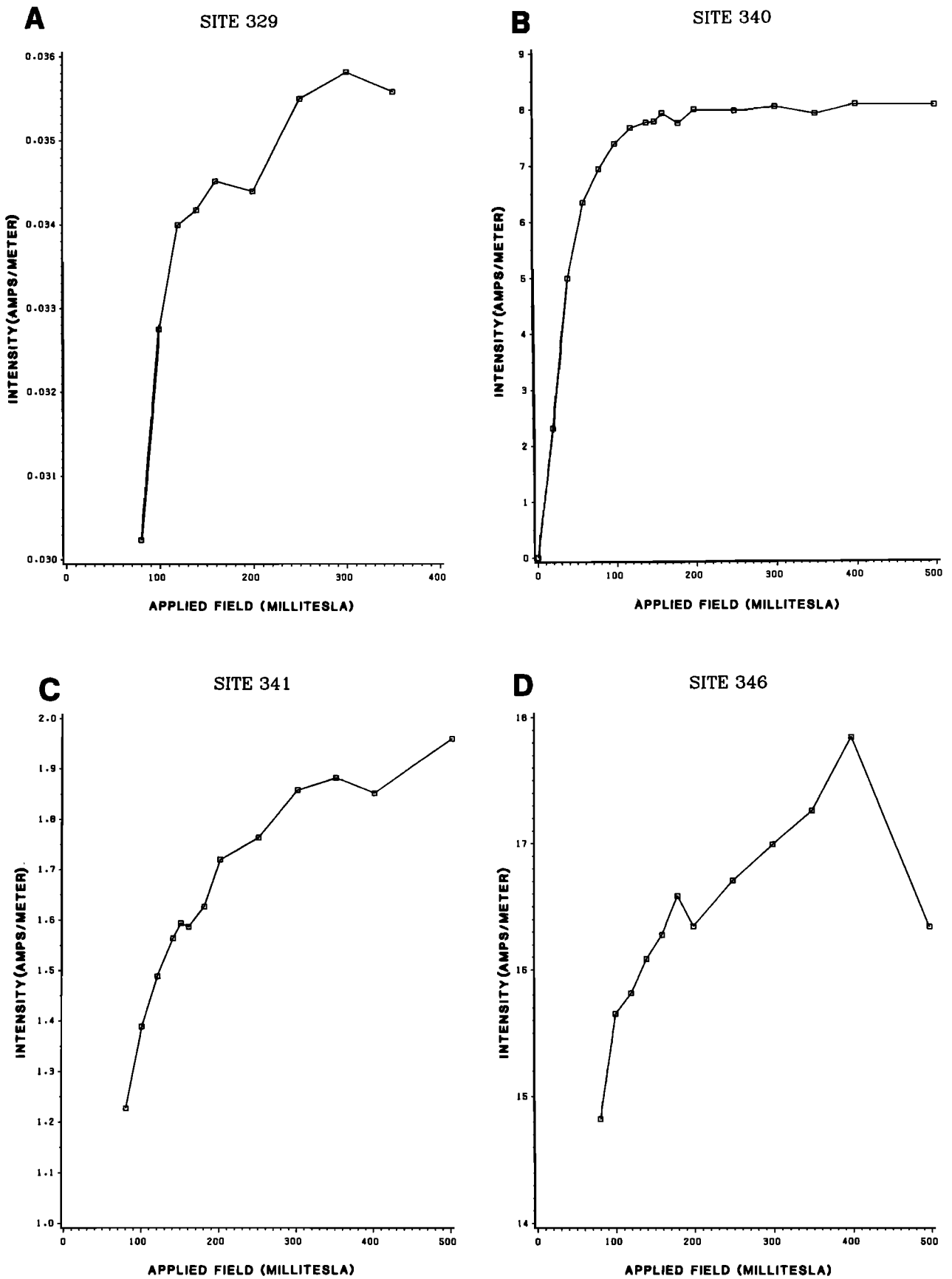


Fig. 4. Plots of acquisition of saturation isothermal magnetization in successively greater applied fields for selected samples from the Hector Formation.

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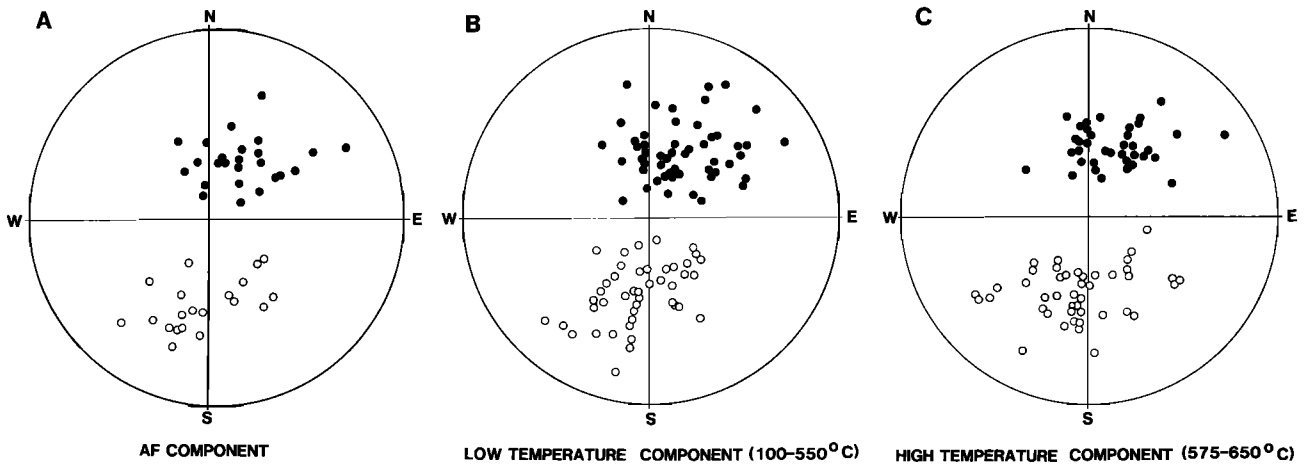


Fig. 5. Stereographic projections of the characteristic directions for the three components of magnetization exhibited for samples from the Hector Formation. Solid symbols are lower hemisphere (positive inclinations) and open symbols are upper hemisphere (negative inclinations). As discussed in text, points with intermediate directions, or inclinations less than ±10° are not used in subsequent data analysis and are consequently not plotted here.

merely constrains the age of magnetization as after 23 m.y., i.e., the age of the bottom of the section.

PALEOMAGNETIC RESULTS

The following steps were taken to select data for structural and geochronological interpretations: (1) screening of individual components, (2) calculation of within-site component statistics, (3) analysis of formational mean statistics for tectonic interpretations, and (4) for interpretation of magnetostratigraphy, the use of site class designation of *Opdyke et al.* [1977].

Data Screening and Selection

Component screening. In almost all cases, paleomagnetic sample vectors for the AF, low temperature, and high temperature components were rejected if maximum angular deviation (MAD) values were >20° after principal component analysis [*Kirschvink*, 1980]. In addition, samples of spurious (e.g., declinations near 0° with negative inclinations) or intermediate (e.g., declinations near 90° or 270° with low [< ±10°] inclinations) directions were rejected. As a result of this screening, 256 principal component directions were choosen for further data interpretation (for AF, N=49; low temperature, N=113; and high temperature, N=94; Table 1, Figure 5). The

TABLE 1. Fisher Parameters for Subsets of Paleomagnetic Data Sets From the Hector Formation

Data Set[a]	D,deg.	I,deg.	N[b]	R	α95	*k*	ΔD
1, samples subjected to AF demagnetization	19.8	46.8	49	44.3	6.7	10.2	9.8
2, low temperature component	16.6	46.5	113	101.0	4.5	9.6	6.5
3, high temperature component	12.8	45.5	94	85.3	4.7	10.7	6.7
4, all sites, tilt corrected	18.6	45.4	48	44.7	5.7	14.0	8.2
5, all sites, in situ	9.0	44.6	48	43.9	6.3	11.6	8.9
6, all normal sites	28.0	45.7	24	22.6	7.4	16.8	10.6
7, all reversed sites	189.3	-44.3	24	22.3	8.3	13.7	11.6

D, declination; I, inclination; R, resultant; α_{95}, 95% cone of confidence; ΔD, declination error margin (see references in text).

[a]Data sets 1-5 include reversed polarity samples inverted to the northern hemisphere. Data sets 1-4, 6, and 7 are tilt corrected.

[b]N, Number of directions used to calculate Fisher statistics. For entries 1-3, this equals the number of individual component directions; for entries 4-7 this equals the number of Class I sites (sensu *Opdyke et al.* [1977]).

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mean directions for each of these three data sets (with reversed sites inverted to northern hemispheres) with their corresponding α_{95} cones of confidence (Figure 6a) overlap each other. This suggests that all three component data sets (1, 2, and 3, in Table 1) are drawn from the same statistical population (i.e., the null hypothesis of no difference cannot be rejected at p=0.05). We therefore argue that this overlap allows these directions, regardless of whether they are from the AF, low, or high temperature components, to be analyzed together within individual sites.

Within-site analysis. The components chosen above for further analysis were next analyzed as individual data points combined within sites using *Fisher* [1953] statistics. For sites with N>4, if there was a component of polarity opposite to the majority of directions, it was deleted from the Fisher analysis. As a result of this statistical analysis, 48 of the 60 originally sampled sites had acceptable results (at p=0.05, for N=3, R>2.62; for N=4, R>3.10; for N=5, R>3.50; for N=6, R>3.85; N=7, R>4.18; N=8, R>4.48; selection criteria taken from *Irving* [1964, Table 4.4]). These Class I sites (sensu *Opdyke et al.* [1977]) were used to calculate the amount of rotation for the Hector Formation.

Site polarity interpretation. In addition to all of the 48 Class I sites, one Class II and two Class III sites were added to the polarity interpretation of the Hector Formation. As defined by *Opdyke et al.* [1977], Class II sites consist of only two available samples, and Class III sites have three samples, only two of which have concordant directions. Therefore, of the 60 originally collected sites, 51 were used for interpretation of a local magnetostratigraphy.

Tests for Stability

The fold test was tried using the statistically significant Fisher site means (Table 1). However, because the structural context of our sampling localities consists of a monocline, a fold test in the strict test, i. e., comparisons of prefolding and postfolding directions in opposite limbs of a given structure [*Graham*, 1949], cannot be employed. The fold test proposed by *McFadden and Jones* [1981], although solving some of the problems with F tests [*McElhinny*, 1964], also can only be employed when there are opposing limbs for comparison. Using the F ratio test for the 48 Fisher site means, the in situ (k_i) versus tilt-corrected (k_t) directions were compared, but the results are inconclusive, because k_i and k_t are not different from one another at p=0.05 (F=1.21).

With regard to other tests for stability, a positive reversals test, with overlapping mean normal and reversed populations (if the α_{95} is considered, Figure 6b), suggests an ancient, stable magnetization and that the demagnetization regime described above effectively isolated characteristic component directions.

Formation Mean Direction and Declination Anomaly

Using the 48 Class I site mean directions presented in Table 2, a Fisher mean direction for the Hector Formation in the northern Cady Mountains was calculated at declination 18.6°, inclination 45.4°. Using the error calculation of *Beck* [1980] (also *Beck et al.* [1986]), ΔD is 8.2°. In order to determine if any trends exist within the section, a least squares regression of tilt-corrected declinations and inclinations versus stratigraphic height for the 48 site means indicates no correlation (at p=0.01), or trend within the section for either declination (R^2=0.07) or inclination (R^2=0.02).

Magnetostratigraphy and Correlation to the Time Scale

Fifty-one sites (48 Class I, 1 Class II, and 2 Class III) were used to interpret a local magnetostratigraphy of the Hector

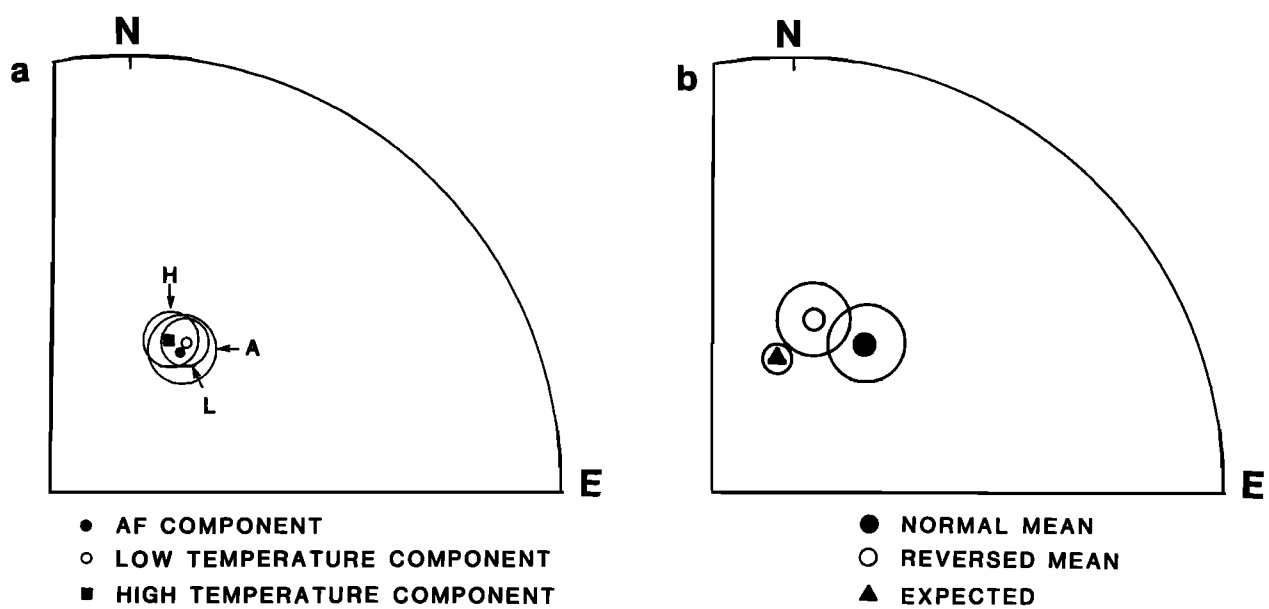


Fig. 6. (a) Plot of mean direction for the AF, low temperature, and high temperature data sets with associated α_{95} circles of confidence. (b) Stereographic projection of Fisher mean declinations and inclinations for all normal and reversed component directions with the associated α_{95} circles of confidence.

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TABLE 2. Fisher Statistics for All Sites (Class I, Tilt-Corrected) Used in Calculation of Rotation

Height	Site	Fisher D, deg	Fisher I, deg	N	R	α_{95}	k
0	340	198.1	-39.7	7	6.89	8.4	52.7
21	342	62.1	42.5	3	2.97	15.0	69.0
27	344	32.8	48.6	6	5.91	8.7	59.7
30	351	22.8	32.9	3	2.97	14.9	69.4
39	349	55.0	23.6	5	4.74	20.0	15.6
41	346	28.5	53.4	5	4.88	13.2	34.4
47	353	44.5	59.8	5	4.93	10.1	58.7
55	348	189.4	-53.0	5	4.57	26.5	9.3
69	357	202.2	-24.5	5	4.96	8.0	92.4
69	359	216.2	-22.8	5	4.82	16.7	22.1
79	350	3.8	55.2	6	5.73	15.9	18.6
85	352	216.8	-28.8	4	3.81	23.6	16.2
90	363	222.5	-51.3	4	3.96	10.1	83.1
93	354	202.3	-38.5	4	3.73	28.8	11.2
102	356	211.9	-32.5	4	3.84	21.9	18.6
110	365	1.3	55.0	7	6.96	5.1	143.3
114	338	27.2	46.9	5	4.50	29.0	7.9
123	324	192.2	-23.8	3	2.87	32.1	15.8
128	331	48.1	50.4	5	4.79	17.8	19.5
134	333	30.2	45.5	7	6.54	17.4	13.0
141	328	190.5	-55.8	4	3.52	40.1	6.2
153	335	20.5	54.0	8	7.67	12.3	21.3
162	330	220.4	-25.7	3	2.97	16.1	59.7
185	339	26.7	40.5	7	6.67	14.7	17.9
194	334	38.1	51.9	5	4.89	12.9	36.1
206	341	48.6	38.6	5	4.80	17.3	20.5
219	336	67.5	30.3	3	2.97	15.2	67.3
222	401	52.1	26.5	3	2.99	6.2	399.2
233	402	193.7	-29.6	5	4.96	7.5	104.2
246	403	17.0	46.1	3	2.95	18.8	44.2
255	404	356.6	48.9	5	4.99	3.1	618.2
257	405	353.8	44.5	5	4.99	4.2	346.5
261	406	28.5	15.4	5	4.97	6.4	143.1
264	422	345.3	41.5	5	4.97	7.2	115.3
273	407	16.9	36.1	4	3.91	16.3	32.9
281	408	352.0	51.7	5	4.97	7.0	120.5
285	409	196.6	-37.5	7	6.96	5.2	136.8
291	410	160.0	-40.2	3	2.98	13.7	81.5
299	411	186.3	-43.3	4	3.99	5.5	277.2
302	412	195.8	-45.4	4	3.59	36.4	7.4
306	413	139.6	-40.2	3	2.91	27.2	21.7
320	415	173.3	-45.0	4	3.77	26.6	12.9
332	416	168.1	-53.2	4	3.95	11.4	66.1
333	417	131.2	-56.3	4	3.99	4.9	349.9
342	418	180.2	-53.0	5	4.93	10.1	59.0
350	419	205.0	-58.5	5	4.87	14.2	30.2
354	420	147.9	-50.0	7	6.93	6.7	82.6
360	421	141.1	-45.0	6	5.81	13.3	26.5
---	all	18.6	45.4	48	44.7	5.7	14.0

The stratigraphic position (height, in meters) and site numbers pertain to Figures 2 and 7.

Formation. These data yielded 15 polarity zones of which five may be questionable because they are only represented by a single site (Figure 7). A unique correlation of this polarity pattern to the standard magnetic polarity time scale (MPTS) is made difficult by two factors: 1) the 17.9 Ma age for the upper part of the section was published without an error margin, and (2) the MPTS for the early and middle Miocene is very "busy," thereby hindering a unique correlation.

Although several correlations to the time scale could be made, the following one seems most plausible if the lower two radioisotopic dates and the paleontological control for the upper part of the section are considered. With regard to the latter aspect, the Cady Mountains Local Fauna is located at the 305-m level. The age of this fauna is late Hemingfordian, which cannot be younger than about 16 Ma [*MacFadden et al.*, 1989]. Given these constraints, the normal polarity zone N1 near the base of the section, which is bracketed by the 22.9 ± 0.4 Ma date, is correlated to C6B-N. Zones R2 to R3 may correlate from C6AA to the bottom of C5E, although the structure of this interval on the MPTS is more complex than observed locally in the Hector Formation. Based on the date of 18.6 ± 0.2 Ma, zones N3 to possibly R6 are correlated to C5E-N to C5D-R on the MPTS. Based on the following three lines of evidence: i.e., (1) the upper paleontological constraint that the section is probably no younger than 16 Ma, (2) the correlations from the underlying zones, and (3) the date of 17.9 Ma, zones N6 to N7 are correlated to the top of C5D on the MPTS and zone R8 is correlated to C5C-R.

It should be noted that zones R5 to R6 are of questionable validity because they are represented each by only a single site; they may correlate to C5D-R (with local zone N5 possibly representing a short event not recognized on the time scale), but this is the least confident part of the correlation proposed here. It alsoneeds to be emphasized here that this is not the only correlation that could be made to the MPTS but is the one that seems most consistent with the associated chronological control. In addition to the three superposed single-site polarity zones (R5 to R6), perhaps the most vexing problem with this correlation is the lack of polarity transitions within the local interval R2-R3 correlated between 22.5 and 19.5 Ma on the MPTS.

REGIONAL TECTONIC INTERPRETATIONS

In order to calculate the amount of rotation for the Hector Formation relative to stable North America, the statistical methods of *Beck* [1980], *Demarest* [1983], and *Beck et al.* [1986] are used. Several late Tertiary poles for North America have been published in the literature (e.g., *Diehl et al.*, 1983); however, we choose that of *Irving and Irving* [1982] because the 20-m.y. mean age is closer to the age of the Hector Formation than the one published by *Diehl et al.* [1983]. Using this paleomagnetic pole [*Irving and Irving*, 1983], the expected declination and inclination for this region during the early Miocene are 358.0° and 53.4°, respectively (α_{95}=4.1°). Taking into account the tilt-corrected Hector formational mean presented above, then the amount of rotation (R) is 20.6° with

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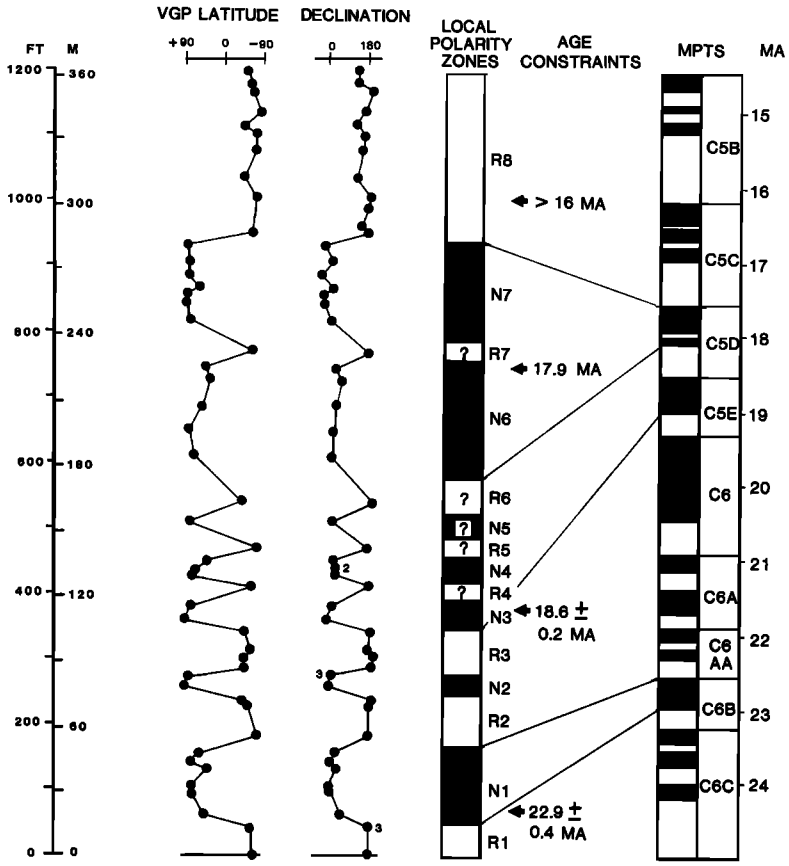


Fig. 7. Paleomagnetic data, chronological control, local magnetic polarity zonation, and correlation to the magnetic polarity time scale (sensu *Berggren et al.* [1985]). The subscripts 2 and 3 next to the declination values represent Class II and Class III sites, respectively (see Figure 2).

an error (ΔR) of 7.6° in a clockwise direction. The flattening (IF ± ΔIF) of 8.0° ± 6.5° and poleward displacement (P ± ΔP) of 1.0° ± 3.4° are considered negligible given the associated error limits.

Because the early Miocene was a time of significant tectonic activity in the Mojave Block, it is possible that some rotation might have occurred in the Cady Mountains during the 7-m.y. time of deposition of the Hector Formation between 23 and 16 Ma. We therefore analyzed the possibility of trends in the clockwise declination anomaly in our stratigraphic section. This would seem a reasonable assumption because other studies have demonstrated Miocene rotations within studied stratigraphic units to the south of the Mojave Block in the Transverse Ranges [*Hornafius et al.*, 1986] and directly to the north of the Garlock fault in the El Paso Mountains [*Loomis and Burbank*, 1988]. However, our least squares regression results presented above indicate that there are no discernable rotational trends during the time of deposition of the Hector Formation.

There are several reports (references cited above) in which about 25 Ma volcanic units in the Mojave Desert have undergone significant rotations (i.e., >25°). It is therefore understandable that the younger sediments of the Hector Formation could have experienced a similar degree of rotation, or less. However, our new results from the northern Cady Mountains are different than those of the the Barstow Formation which is of similar age to the upper part of the Hector Formation [*MacFadden et al.* 1987, 1990]. Located some 50 km to the west (Figure 1), the Barstow sediments of the Mud Hills indicate negligible or slight counterclockwise rotation after 13 Ma. The explanation for this difference in contemporaneous sediments may be related to the presence of the Hector and Barstow formations in different structural settings, i.e., bounded by different principal fault trends (E-W and NW, respectively). As suggested by *Ross et al.* [1989], the currently available data base now indicates at least two principal periods of rotation; the first in the early Miocene (about 25 Ma), associated with the earliest phase of regional extension, and a second during the middle Miocene, or later, and is possibly related to movement along the San Andreas fault [*Ross et al.*, 1988; *Dokka and Travis*, 1990]. The age of the second phase of tectonic rotation is difficult to constrain within the Mojave Block because there is a major gap between early Miocene units and younger ones with associated paleomagnetic data. The Black Mountain Basalt, located to the northwest of the Mud Hills, is dated at 2.55 ± 0.58 Ma and appears not to be rotated [*Burke et al.*, 1982]. Although this determination may represent an instantaneous pole, previous

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structural models also predict little rotation in this part of the Mojave Block.

Our data are also in marked contrast to much larger magnitudes of clockwise rotations reported in adjacent structural terranes outside of the Mojave Block. For example, rocks on the outboard side of the San Andreas Fault zone have experienced large Neogene rotations, e.g., as much as 100° and 41.4° ± 7.7° (clockwise) from the western and eastern Transverse Range, respectively (Figure 1) [*Hornafius et al.*, 1986; *Carter et al.* 1987] (see review by *Luyendyk and Hornafius* [1987]). Similarly, in the El Paso Basin (Figure 1) to the north of the Mojave Block, *Loomis and Burbank* [1988] (also see *Burbank and Whistler* [1987]) analyzed the paleomagnetism of the Ricardo Group. Their results indicate significant counterclockwise rotation that occurred within the deposition of this unit between about 13 and 7 Ma that is probably related to left-lateral movement along the Garlock fault zone.

Although the tectonic models of *Garfunkel* [1974] and *Luyendyk et al.* [1980] for the Mojave Desert are insightful, it is clear from the expanding paleomagnetic data base that the entire region cannot be explained by a simple model of uniform rotation during the middle to late Cenozoic. Each individual block must be evaluated for their demonstrated rotational histories before any regional pattern can emerge.

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B. J. MacFadden, Florida Museum of Natural History, University of Florida, Gainesville, FL 32611.

N. D. Opdyke, Department of Geology, University of Florida, Gainesville, FL 32611.

M. O. Woodburne, Department of Geological Sciences, University of California, Riverside, CA 92521.

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